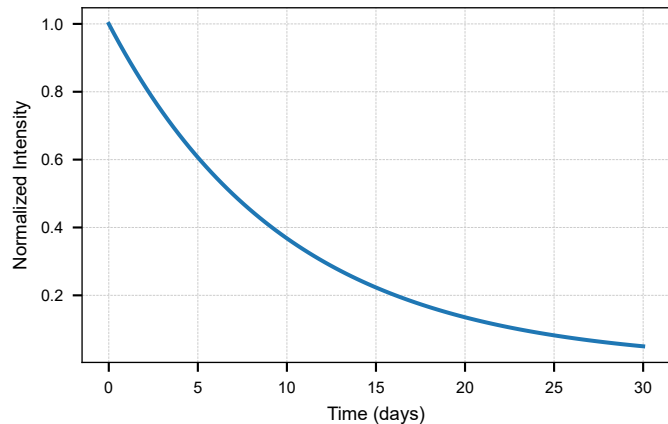
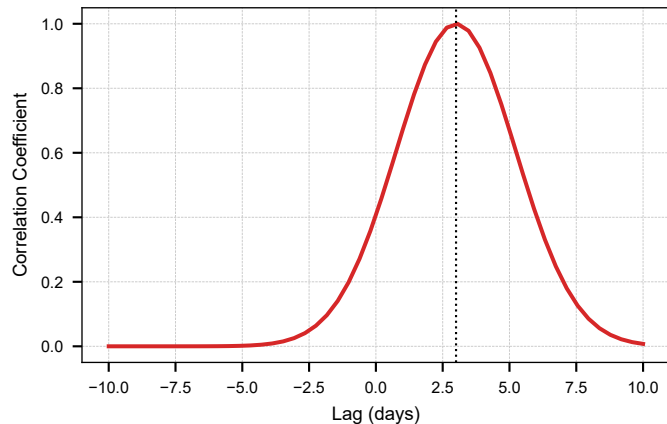


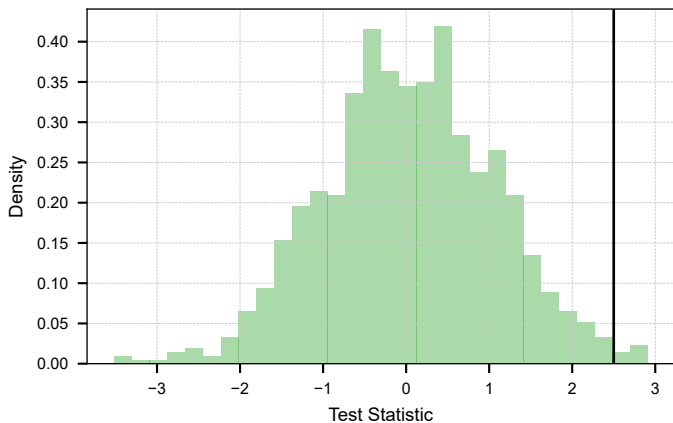
(a) Satellite Disappearance Signal



(b) Cross-Correlation Lag



(c) Null Model Distribution



(d) Validation Workflow Summary

Lag Correlation

$r = 0.84$ ($p < 0.001$)

Status: Passed ✓

Granger Causality

$F = 14.2$ ($p = 0.002$)

Status: Passed ✓

Null Significance

$p_{\text{null}} < 0.005$

Status: Passed ✓